



P. 21

$$\text{conduction loss} = i_{\text{rms}}^2 \cdot R_{\text{ds-on}}$$

$$\text{switching loss} = \text{const}$$

$$\text{driving loss} = \text{const}$$

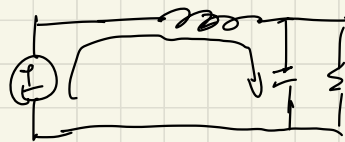
In heavy-load mode, conduction loss is dominant

$$\rightarrow \eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{loss}}} \quad \begin{array}{l} P_{\text{out}} \propto i_{\text{rms}} \\ P_{\text{loss}} \propto i_{\text{rms}}^2 \end{array}$$

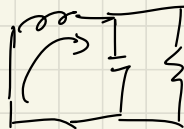
$$i_{\text{rms}} \uparrow \quad P_{\text{loss}} \uparrow \quad \eta \downarrow$$

~~*~~

p. 32



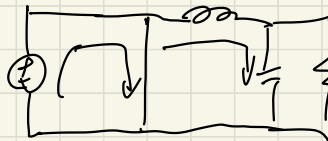
reverse bias



forward bias

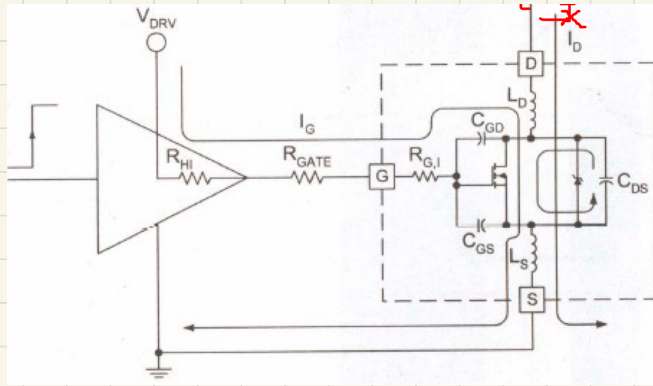
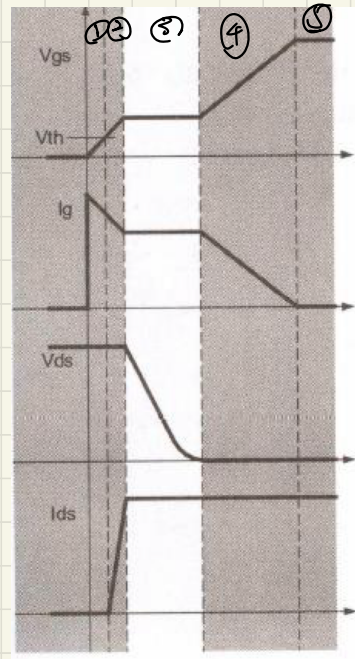


③



reverse recovery happens
at the transition between ③ and ①

P.52



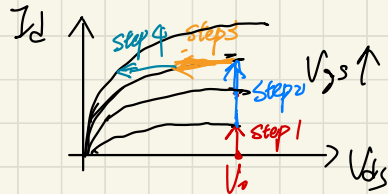
① V_{gs} start to rise (discharge C_{GD} , C_{GS})

② $V_{gs} = V_{th}$ MOS turns on, I_{DS} starts rising

③ After $I_{DS} = I_o$, diode clamp disappears, V_{ds} falls

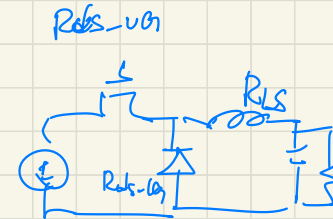
④ V_{ds} falls to conducting condition, V_{gs} starts to rise

MOS operation curve at $I_{ds} - V_{ds}$ map



2. (70%) P. 22 shows an open-loop NN-power stage buck converter with specified circuit parameters.

- (a) (30%) Please derive the equations of R_o of the equivalent circuit in p. 23 and the power efficiency as a function of circuit parameters. Then substitute parameter values into the derived equations to get R_o and efficiency values.
- Use circuit simulation to verify the calculation results and reproduce the figures in p. 23, 24. Suggested simulation tool: LTspice or Virtuoso
- (b) (40%) Please implement a driver circuit with functions of (1) Drive both switches properly. (2) Dead-time control to avoid shoot-through (3) Zero current detection for DCM operation.
- The functions can be implemented by behavioral circuits and tested by providing a given duty signal.
- (1) Show the implemented driver and switch circuit. Explain its operation. (2) Show the simulated waveforms including duty, gate drive signals v_{UG_DRIVE} and v_{LG_DRIVE} , V_{LX} (VSW), i_L to prove the circuit achieves its function.
- You got maximum 15 extra point if you can proof that your driver implementation have any advantage compared to the fixed dead-time, comparator ZCD detection, conventional level shifter, and bootstrap circuit in lecture notes.

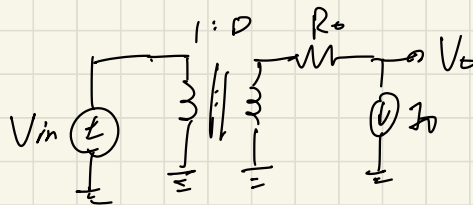


$$\begin{aligned} & \rightarrow (V_{in} - I_o R_{ds(on)} - I_o R_{DS} - V_o) \cdot D \\ & + (-I_o R_{ds(on)} - I_o R_{DS} - V_o) (1-D) \approx 0 \end{aligned}$$

(a)

$$V_o = D_{ideal} V_{in} - I_o \left[\underbrace{D_{ideal} R_{ds-on}}_{\substack{\downarrow \\ \text{ON-time } R_{ds-on}}} + \underbrace{(1 - D_{ideal}) R_{ds-on}}_{\substack{\downarrow R_o \\ \text{OFF-time } R_{ds-on}}} + \underbrace{R_{Ls}}_{\substack{\rightarrow \text{inductance resistance} \\ (R_{ds-on} + R_{Ls})}} \right]$$

equivalent circuit

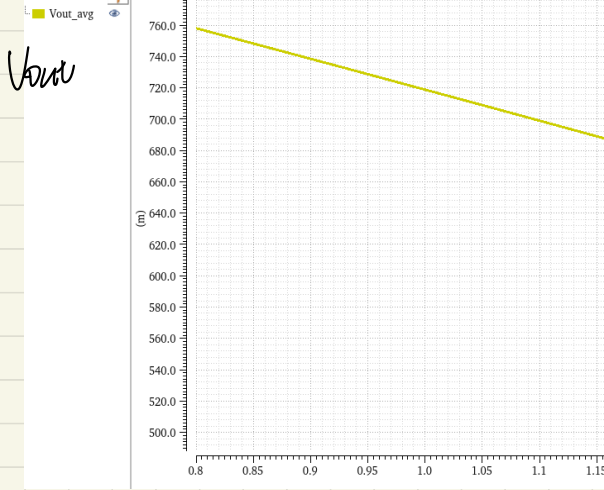


$$\text{Efficiency } \eta = \frac{P_{out}}{P_{in}} = \frac{V_o \cdot I_o}{V_{in} \cdot D I_o} = \frac{V_o}{D V_{in}} = \frac{D V_{in} - I_o R_o}{D V_{in}}$$

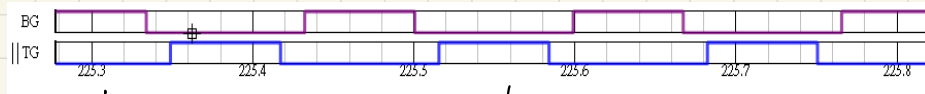
$$R_o = D R_{ds-on} + (1 - D) R_{ds-on} + R_{Ls}$$

$$R_o = 0.5 \cdot 100m + 0.5 \cdot 100m + 100m = 200m\Omega$$

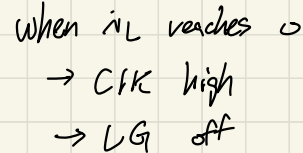
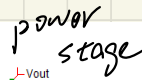
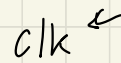
$$\text{efficiency} = \frac{0.8 - 200m I_o}{0.8} \quad \#$$



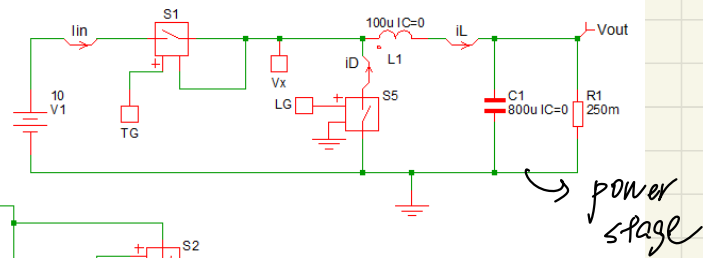
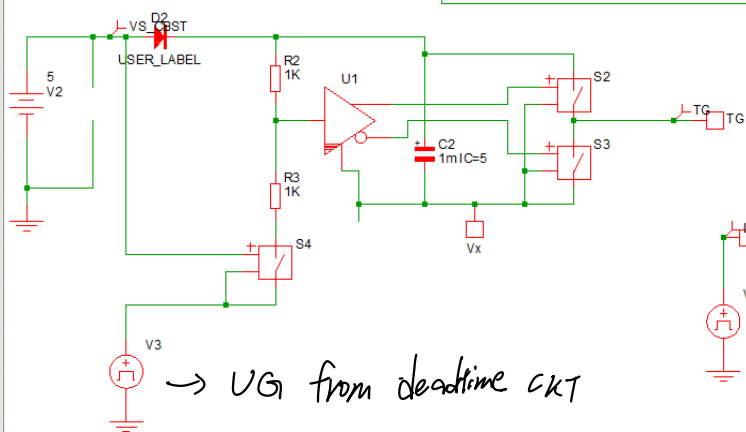
fix deadtime circuit



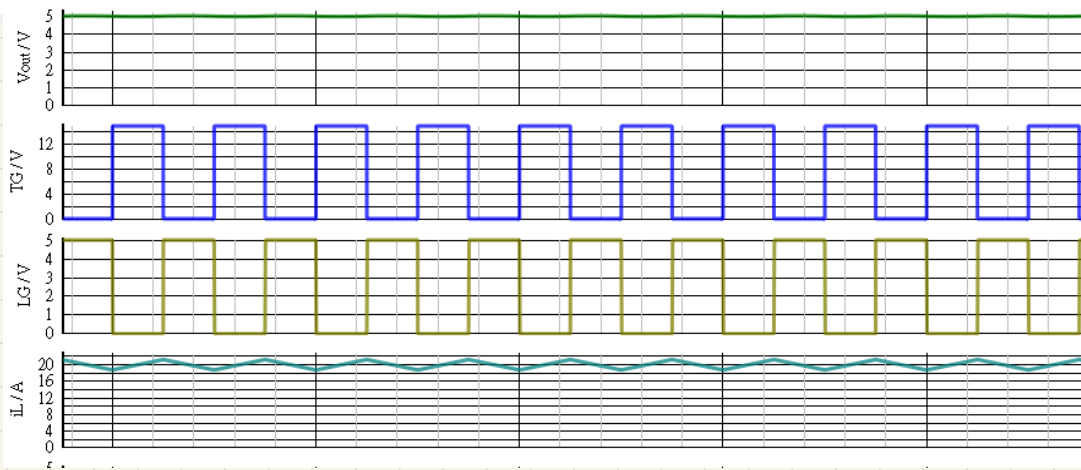
make sure BG TG has deadline (20ns)



LSF



LG from ZCD



V_{UG_drive}

VLOG DRIVE

 $\hat{A}_L$

3. (5%) PMIC for AIoT System (see next page). To achieve 85% efficiency at $I_o = 0.1 \text{ mA}$, what is the maximum controller quiescent current budget? Hint: assume the converter loss is dominated by controller quiescent current.

$$0.85 = \frac{P_{out}}{P_{out} + P_{loss}} = \frac{0.1 \cdot V_o}{0.1 \cdot V_o + I_q \cdot V_o} = \frac{0.1}{0.1 + I_q} = 0.85$$

$$0.085 + 0.85 I_q = 0.1$$

$$0.015 = 0.85 I_q$$

$$I_q = 0.0176 \text{ mA} \\ = 17.6 \mu\text{A}$$